Year 11 · AQA Level 2 Further Maths · Covered topic

Expanding and factorising, including three brackets

Revisit and practise · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Expand products of two and three brackets

2. Factorise harder expressions and quadratics

Do now: retrieval practice

4 mins

1 Expand $3(2x - 5)$

$6x - 15$

2 Factorise $12x + 18$

$6(2x + 3)$

3 Expand $(x + 4)(x - 2)$

$x^2 + 2x - 8$

4 Factorise $x^2 - 9$

$(x - 3)(x + 3)$

Objective 1

Expand products of two and three brackets

Key idea

Learn

!

Multiply every term in one bracket by every term in the other. For three brackets, expand two first, then multiply the answer by the third bracket. Collect like terms at the end.

Objective 1: Expand products of two and three brackets

I do: example 1

I do

Expand $(2x + 3)(x - 4)$

1. $2x \times x = 2x^2$ and $2x \times (-4) = -8x$
2. $3 \times x = 3x$ and $3 \times (-4) = -12$
3. $2x^2 - 8x + 3x - 12$
4. $= 2x^2 - 5x - 12$

I do: example 2

I do

Expand $(x + 1)(x + 2)(x + 3)$

1. $(x + 1)(x + 2) = x^2 + 3x + 2$
2. Multiply by $(x + 3)$
3. $x^3 + 3x^2 + 3x^2 + 9x + 2x + 6$
4. $= x^3 + 6x^2 + 11x + 6$

I do: example 3

I do

Expand $(x - 2)^3$

1. $(x - 2)(x - 2) = x^2 - 4x + 4$
2. Multiply by $(x - 2)$
3. $x^3 - 2x^2 - 4x^2 + 8x + 4x - 8$
4. $= x^3 - 6x^2 + 12x - 8$

Mini whiteboards

We do

1. Expand $(3x - 1)(2x + 5)$

Answer: $6x^2 + 13x - 5$

2. Expand $(x + 2)(x - 1)(x + 3)$

Answer: $x^3 + 4x^2 + x - 6$

3. Expand $(x - 4)(2x + 3)$

Answer: $2x^2 - 5x - 12$

4. Expand $(x + 1)^3$

Answer: $x^3 + 3x^2 + 3x + 1$

You do: choose your challenge

You do

Bronze

Expand $(x + 5)(x - 3)$

Silver

Expand $(2x - 3)^2$

Gold

Expand $(x + 3)(x - 1)(x - 2)$

Platinum

Expand $(2x + 1)^3$

You do: answers

Check

Bronze

$x^2 + 2x - 15$

Silver

$4x^2 - 12x + 9$

Gold

$x^3 - 7x + 6$

Platinum

$8x^3 + 12x^2 + 6x + 1$

If you finish early

Extra

Bronze

Expand $(x + 6)(x - 2)$

$x^2 + 4x - 12$

Silver

Expand $(3x + 2)^2$

$9x^2 + 12x + 4$

Gold

Expand $(x + 2)(x + 3)(x - 4)$

$x^3 + x^2 - 14x - 24$

Platinum

Expand $(x - 1)(x + 1)(2x + 3)$

$2x^3 + 3x^2 - 2x - 3$

Exam question 1

3 marks

Expand and simplify $(2x + 1)(x - 3)(x + 2)$

Space for working

Mark scheme: $(2x + 1)(x - 3) = 2x^2 - 5x - 3$ (M1). Multiply by $(x + 2)$ (M1). $2x^3 - x^2 - 13x - 6$ (A1)

Exam question 2

2 marks

Show that $(x + 3)^2 - (x - 3)^2 = 12x$

Space for working

Mark scheme: $x^2 + 6x + 9 - (x^2 - 6x + 9)$ (M1) $= 12x$ (A1)

Objective 1 achieved

Expand products of two and three brackets



Objective 2

Factorise harder expressions and quadratics

✓ Achieved: Expand products of two and three brackets

Key idea

Learn

!

Always take out the highest common factor first. For $ax^2 + bx + c$, find two numbers that multiply to ac and add to b, split the middle term, then factorise in pairs. Spot the difference of two squares: $a^2 - b^2 = (a - b)(a + b)$.

Objective 2: Factorise harder expressions and quadratics

I do: example 1

I do

Factorise $6x^2 + 7x - 3$

1. $ac = 6 \times (-3) = -18$: the numbers are 9 and −2
2. $6x^2 + 9x - 2x - 3$
3. $3x(2x + 3) - 1(2x + 3)$
4. $= (3x - 1)(2x + 3)$

I do: example 2

I do

Factorise fully $2x^3 - 18x$

1. Take out 2x: $2x(x^2 - 9)$
2. $x^2 - 9$ is a difference of two squares
3. $= 2x(x - 3)(x + 3)$

I do: example 3

I do

Factorise fully $4x^2y - 12xy^2$

1. HCF of 4 and 12 is 4
2. HCF of x^2y and xy^2 is xy
3. $= 4xy(x - 3y)$

Mini whiteboards

We do

1. Factorise $3x^2 - 10x - 8$

Answer: $(3x + 2)(x - 4)$

2. Factorise $9x^2 - 25y^2$

Answer: $(3x - 5y)(3x + 5y)$

3. Factorise $2x^2 + 9x + 4$

Answer: $(2x + 1)(x + 4)$

4. Factorise fully $x^3 - 4x$

Answer: $x(x - 2)(x + 2)$

You do: choose your challenge

You do

Bronze

Factorise $x^2 - 11x + 28$

Silver

Factorise $2x^2 + 5x + 3$

Gold

Factorise fully $3x^3 - 12x$

Platinum

Factorise $6x^2 - x - 12$

You do: answers

Check

Bronze

$(x - 4)(x - 7)$

Silver

$(2x + 3)(x + 1)$

Gold

$3x(x - 2)(x + 2)$

Platinum

$(3x + 4)(2x - 3)$

If you finish early

Extra

Bronze

Factorise $x^2 + 2x - 35$

$(x + 7)(x - 5)$

Silver

Factorise $3x^2 + 7x + 2$

$(3x + 1)(x + 2)$

Gold

Factorise fully $50x^2 - 8$

$2(5x - 2)(5x + 2)$

Platinum

Factorise $10x^2 - 11x - 6$

$(5x + 2)(2x - 3)$

Exam question 1

2 marks

Factorise fully $5x^2 - 45$

Space for working

Mark scheme: $5(x^2 - 9)$ (M1) $= 5(x - 3)(x + 3)$ (A1)

Exam question 2

2 marks

Factorise $4x^2 + 4x - 15$

Space for working

Mark scheme: (2x + 5)(2x − 3) (B2)

Objective 2 achieved

Factorise harder expressions and quadratics



Real life

Apply

A box is x cm wide, (x + 2) cm long and (x − 1) cm tall. Write an expression for its volume.

$x(x + 2)(x - 1) = x^3 + x^2 - 2x$ cm^3

Extension: stretch yourself

Stretch

1. Expand $(x + a)^3$

$x^3 + 3ax^2 + 3a^2x + a^3$

2. Factorise $x^4 - 16$ fully

$(x^2 + 4)(x - 2)(x + 2)$

Exit ticket: exam questions

Exit

1. Expand and simplify $(x + 4)(x - 1)(x + 2)$ **(3 marks)**

2. Factorise $3x^2 + 11x + 6$ **(2 marks)**

3. Factorise fully $2x^2 - 50$ **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $(x^2 + 3x - 4)(x + 2)$ (M1) $= x^3 + 5x^2 + 2x - 8$ (A2) [3]

Q2 $(3x + 2)(x + 3)$ [2]

Q3 $2(x - 5)(x + 5)$ [2]

Total: ____ / 7

Today I achieved

- ✔ **Expand products of two and three brackets**
- ✔ **Factorise harder expressions and quadratics**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Expand and simplify $(3x + 2)(2x - 5)$ [2]

2. Expand and simplify $(x - 1)(x + 2)(x + 5)$ [3]

3. Factorise $5x^2 + 13x - 6$ [2]

4. Factorise fully $12x^2 - 27$ [2]

5. Show that $(n + 1)^2 - (n - 1)^2$ is a multiple of 4 for all integers n [3]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Expanding three brackets"